HINT & SOLUTION OF FLUID MECHANICS

EXERCISE-1

PART - I

A-1. Sharp kinife applies more pressure as compare to blunt knife because of lesser area of contact.

A-2. It has high specific gravity.

A-3. $p = h\rho g$

$15 \times 10^3 = 3 \times \rho \times 10$

$\rho = 500 \text{ kg/m}^3$

$s = \dfrac{\rho}{\rho_w} = \dfrac{500}{1000} = 0.5$

A-4. $P_x + \rho_\omega g\left(\dfrac{175}{100}\right) - \rho_{Hg} g\left(\dfrac{112}{100}\right) + \rho_\omega g\left(\dfrac{75}{100}\right) - \rho_{Hg} g\left(\dfrac{88}{100}\right) - \rho_\omega g\left(\dfrac{62}{100}\right) = P_y$

Solving $P_x - P_y = 253200 \text{ N/m}^2$ $(g = 10 \text{ m/sec}^2)$

A-5. Initially $\left[\dfrac{0+\rho ga}{2}\right](a)^2 + \left[\dfrac{\rho ga + 3\rho ga}{2}\right](a^2) = 2.5\ \rho ga^3$

Finally $\left[\dfrac{0+\rho ga}{2}\right](2a^2) = 2\rho ga^3$

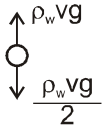
Difference $= 0.5\ \rho ga^3$ [Which is one fifth of initial]

B-1. $\rho(a^2)(2) = 200 \text{ g}$

but $\rho = 1 \text{ gm/cm}^3$

$a = 10 \text{ cm}$.

B-2. In water $a = g$



So, by symmetry in air and water in acceleration distance travelled = 19.6 m.

$t = 2\sqrt{2h/g} = 2\sqrt{2 \times 19.6/9.8} = 4$ sec.

B-3. Till the top edge reaches the surface of water. Tension = weight – buoyant force and buoyant force till then is constant. When plate is partially in water and partially in air, then as plate move up buoyant force decreases and when plate comes out tension again becomes constant or T = Weight.

C-1. $A_1v_1 = A_2v_2$ .....(i)

$P_1 + \dfrac{1}{2}\rho v_1^2 + 0 = P_2 + \dfrac{1}{2}\rho v_2^2 + 0$ .....(ii)

Given, $P_1 - P_2 = 10 \text{ N/m}^2$

$\rho = 1.25 \times 10^3$

$r_1 = 0.1 \text{ m}$

$r_2 = 0.04 \text{ m}$

Solving, rate of flow of glycerin $= A_1V_1 = A_2V_2 = 6.43 \times 10^{-4} \text{ m}^3/\text{sec}$.

C-2. $av' = Av$
$A = 5a$
$v' = 5v$
$P_A + \frac{1}{2}\rho v_A^2 + 0 = P_a + \frac{1}{2}\rho v_a^2$
but $P_a = 0$ and $P_A = 2$ atm
Solving $Av \equiv 8.1 \times 10^{-3}$ m^3/sec.

C-3. $A_1v_1 = A_2v_2$
(i) 10 cm^3/s = 40 mm^2 × v_x or $v_x = 25$ cm/s
(ii) 10 cm^3/s = 20 mm^2 × v_y or $v_y = 50$ cm/s

$$P_x + \frac{1}{2}\rho v_x^2 = P_y + \frac{1}{2}\rho v_y^2$$

$$P_x - P_y = \frac{1}{2}\rho v_y^2 - \frac{1}{2}\rho v_x^2 = \frac{1}{2} \times 1000 \left[\left(\frac{1}{2}\right)^2 - \left(\frac{1}{4}\right)^2\right]$$

or $P_x - P_y = 93.75$ N/m^2 .

C-4. $A_1v_1 = A_2v_2$

$$P_y + \frac{1}{2}\rho v_y^2 + 0 = P_x + \frac{1}{2}\rho v_y^2 + \rho gh.$$

$$P_x - P_y = \frac{1}{2}\rho(v_y^2 - v_x^2) - \rho gh$$

$$P_x - P_y = \frac{1}{2}1000\left[\left(\frac{1}{2}\right)^2 - \left(\frac{1}{4}\right)^2\right] - 1000 \times 10 \times \frac{15}{1600}$$

$$P_x - P_y = 0$$

C-5. $A_1v_1 = A_2v_2$

$$P_y + \frac{1}{2}\rho v_y^2 + \rho gh = P_x + \frac{1}{2}\rho v_x^2 + 0.$$

$$P_x - P_y = \frac{1}{2}\rho(v_y^2 - v_x^2) + \rho gh$$

$$P_x - P_y = \frac{1}{2}1000\left[\left(\frac{1}{2}\right)^2 - \left(\frac{1}{4}\right)^2\right] + 1000 \times 10 \times \frac{15}{1600}$$

$P_x - P_y = 187.5$ N/m^2

C-6. through Bernoulli's equation
If pressure becomes zero, then velocity will be maximum

$$P_{atm} + 0 = 0 + \frac{1}{2}\rho v_{max}^2 \qquad v_{max} = \left(\frac{2p_{atm}}{\rho}\right)^{1/2}$$

PART - II

A-1. $\rho g(H - h)$
Thrust is equal to pressure at that point.

A-2. $F = [\rho gh]\,[A]$
$= (1000)\,(10)\,(6)\,(10)\,(8)$.

A-3. $\frac{m_1 g}{A_1} = \frac{m_2 g}{A_2}$

Solving $m_2 = 3.75$ kg.

A-4. $W_A > W_B$ as mass of water in A is more than in B

$P_A = P_B$

Area of A = Area of B

or $P_A \text{ Area}_A = P_B \text{ Area}_B$

or $F_A = F_B$.

A-5. (i) $a = a_0 (\hat{i} - \hat{j} + \hat{k})$

As there is no gravity; the pressure difference will be only due to the acceleration.

At point B the pseudo force is maximum hence pressure is maximum.

(ii) At point H the pseudo force is minimum hence pressure is minimum

B-1. $\Delta v = v_f - v_i = \frac{m}{y} - \frac{m}{x}$

B-2. $mg = 60$ .....(i)

$mg - \rho_i vg = 40$ .....(ii)

$\frac{mg - \rho_\ell vg}{mg} = \frac{2}{3}$ or $\frac{\rho_0}{\rho_\ell} = 3$

where ρ_0 = density of the block and ρ_ℓ = density of the liquid.

B-3. $\Delta U = mgh$

$\Delta U = \sigma_b Vgh$

B-4. $W_{app} = mg - F_B$

$W_{app} = \rho Vg - \rho_w Vg$

$= (\rho - \rho_w) Vg$

$= (7\rho_w - \rho_w) Vg = 6 \rho_w Vg$

B-5. $10^3 \times \frac{4}{5} + 13.5 \times 10^3 \times \frac{1}{5} = \rho \times 1$

or $\rho = 3.5 \times 10^3$ kg/m^3

B-6. $[36 - \rho_\ell v_1]g = [48 - \rho_\ell v_2]g$

$$\left[36 - \rho_\ell \left(\frac{36}{9}\right)\right] g = \left[48 - \rho_\ell \left(\frac{48}{\rho_0}\right)\right] g$$

Solving

$\rho_0 = 3$.

B-7. In stable equilibrium the object comes to its original state if disturbed.

B-8. As, weight = Buoyant force

$mg = [100 \times 6 \times 0.6\, g] + (100 \times 1 \times 4)g$ $\therefore$ $m = 760$ gm.

C-1. $R = vt$

$= \sqrt{2gD}\ \sqrt{\frac{2(H-D)}{g}}$

$= 2\sqrt{D(H-D)}$.

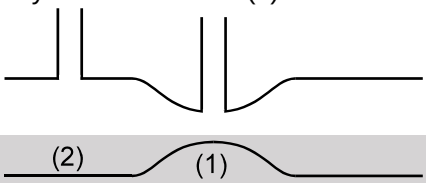
C-2. $x = 2\sqrt{h(H-h)}$

for x_{max} $\frac{dx}{dh} = 0$ or $h = \frac{H}{2}$

C-3. from equation of continuity

$(A \times 3) = (A \times 1.5) + (1.5\ A \times V) \quad \Rightarrow \quad V = 1\ m/s^2$

C-4. From continuity equation, velocity at cross-section (1) is more than that at cross-section (2).



Hence; $P_1 < P_2$

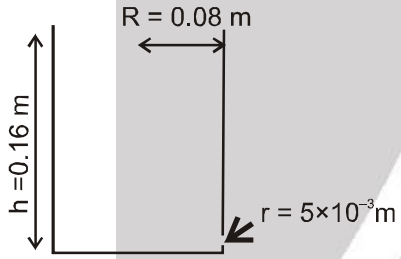
C-5. $F_{thrust} = \rho a v^2$

$F_{net} = F_1 - F_2 = a\rho[2g(h_1 - h_2)]$

$= a\rho(2gh)$

or $F \propto h$

C-6.



$A_1 v_1 = A_2 v_2$

$\pi R^2\ dh/dt = \pi r^2\ v$ ....(i)

$v = \sqrt{2gh}$ ....(ii)

from equation (ii) put the value of v in equation (i)

$\pi R^2\ dh/dt = \pi r^2\ \sqrt{2gh}$

$\Rightarrow \quad \int \frac{R^2 dh}{r^2\sqrt{2gh}} = \int dt$

$\frac{R^2}{r^2\sqrt{2g}} \int_h^0 \frac{dh}{\sqrt{h}} = \int_0^T dt$

$T = \frac{R^2}{r^2}\sqrt{\frac{2h}{g}}$

on solving t = 46.26 second.

C-7. Thrust on curved surface $= < p > 2\pi RH$

$= 15\ .\ 2\pi R(10)$

Force on bottom

$= < p > \pi R^2 = 20\ .\pi R^2$

$15\ 2\pi R(10) = 20\ \pi R^2$

$\therefore R = 15 m$

C-8. Thrust force

$F = \rho A v_{rel}^2$

$v_{rel} = (v - V)$

$\therefore\ F \propto v_{rel}^2$

PART - III

1. Pressure varies with height $\Rightarrow P = \rho gh$
and is horizontal with acceleration $\Rightarrow P = \rho \ell a$
so on (A) ρgh part is zero while average force of ρax is

$\left[\frac{0+\rho\ell a}{2}\right]$ $[\ell^2]$

$= \frac{\ell \rho a}{2}(\ell^2) = \frac{(\rho \ell^3)}{2}\ a = \frac{ma}{2}$

In (B) $\rho \ell a$ part is zero while average force of ρgx is

$\left[\frac{0+\rho g\ell}{2}\right]\ \left[\ell^2\right] = \frac{\rho g}{2}\ (\ell^3)$

$= \frac{\rho(\ell^3)}{2}(g) = \frac{mg}{2}$

Similarly for other part

2. (A) On ABCD avg pressure $= \left[\frac{0+\rho_1 gh}{2}\right]$

(A) ABCD $= \left[\frac{0+\rho_1 gh}{2}\right]$ So F $= \left[\frac{\rho_1 gh}{2}\right]$ $[\ell h] = \frac{\rho_1 gh^2 \ell}{2}$

(B) No contact of ρ_2 and not any pressure on ABCD due to ρ_2
(C) On CDEF due to ρ_1, at every point pressure is $\rho_1 gh$ so average is also $\rho_1 gh$
so F = $(\rho_1 gh)$ $(h\ell) = \rho_1 gh^2 \ell$

(D) On CDEF force due to liquid of density ρ_2 is $\frac{\left[\rho_2 gh^2 \ell\right]}{2}$

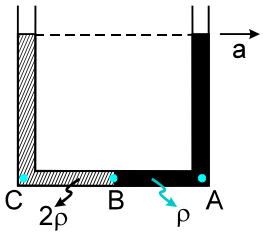
EXERCISE - 2

PART - I

1. $F_{th} = \frac{\sqrt{2}dp}{dt} = \sqrt{2}V\frac{dm}{dt} = \sqrt{2}V\ [\rho\ L\ ] = \sqrt{2}\rho VL$

2. Pressure exerted by fluid at closed end B is $P = \rho g\ell$
$\therefore$ Force exerted by fluid at closed end B is $F = PA = \ell \rho g\ A_0$

3. For the given situation, liquid of density 2ρ should be behind that of ρ.



From right limb :
$P_A = P_{atm} + \rho\, gh$

$P_B = P_A + \rho a\frac{\ell}{2} = P_{atm} + \rho gh + \rho a\frac{\ell}{2}$

$P_C = P_B + (2\rho) a\frac{\ell}{2} = P_{atm} + \rho gh + \frac{3}{2}\rho a\ell$ .... (1)

But from left limb :

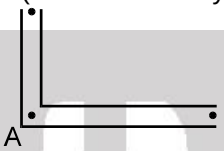
$P_C = P_{atm} + (2\rho) gh$ .... (2)

From (1) and (2) :

$P_{atm} + \frac{3}{2}\rho gh + \rho a\ell = P_{atm} + 2\rho gh \quad \Rightarrow \quad h = \frac{3a}{2g}\ell$ **Ans.**

4. No sliding $\Rightarrow$ pure rolling

Therefore, acceleration of the tube = 2a (since COM of cylinders are moving at 'a')

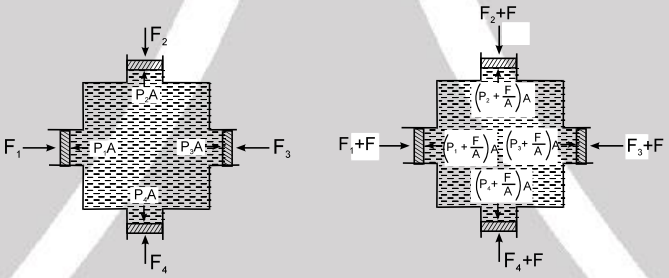


$P_A = P_{atm} + \rho (2a) L$ (From horizontal limb)

Also ; $P_A = P_{atm} + \rho g H$ (From vertical limb) $\Rightarrow$ $a = \frac{gH}{2L}$ **Ans.**

5. (B) As long as $\rho \le \rho_W$, pressure at the bottom of the pan would be same everywhere, according to the Pascal's law.

6. The four piston are initially in equilibrium. As additional force F is applied to each piston, the pressure in fluid at each point must be increased by $\frac{F}{A}$ so that each piston retains state of equilibrium.



Thus the increment in pressure at each point is $\Delta P = F/A$ (by Pascal's law)

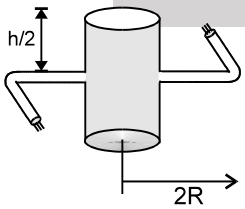
7. Since not touching,

So, $R = F_b = \rho_l(vg) = 40g$.

$R' - R = 80g - 40g = 40g$

Hence R′ will be 40g more than R

8.



Velocity of efflux of water (v) = $\sqrt{2g\left(\frac{h}{2}\right)} = \sqrt{gh}$

Force on ejected water = Rate of change of momentum of ejected water.

$= \rho (av) (v) = \rho av^2$

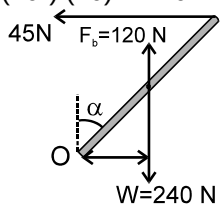
Torque of these forces about central line

$= (\rho av^2) 2R.2 = 4\rho av^2 R = 4\rho agh R$

9. F.B.D. of rod

$W = (0.012)\ (1)\ (2 \times 10^3)\ (10) = 240$ N

$F_b = (0.012)\ (1)\ (10^3)\ (10) = 120$ N



Torque about O

(For equilibrium

$(240 - 120)\left(\frac{\sin\alpha}{2}\right) = 45\ (\cos\alpha)$

$\Rightarrow \tan\alpha = \frac{90}{120} = \frac{3}{4} \Rightarrow \alpha = 37°$

10. Torque about CM

$F_b \cdot \frac{\ell}{4} = I\alpha$

$\Rightarrow \qquad \alpha = \frac{1}{I}(\pi r^2)\ (\ell)\ (\rho)\ (g).\ \frac{\ell}{4} \qquad \alpha = \frac{\pi r^2 \ell^2 g\rho}{4I}$

'α' will be same for all points on cylinder

Hence (B).

11. Increasing the temperature of water from 2^0C to 3^0C increases its density while decreases the density of iron.

Hence the bouyant force increases.

12. $y = \frac{\omega^2 r^2}{2g}$

Put values and get y = 2cm.

13. Buoyant force = $F_b = V_{sub} \cdot \rho_\ell \cdot g$

where, V_{sub}, ρ_ℓ and g all are same w.r.t. O_1 and O_2.

Hence (A)

14. $A_1V_1 = A_2V_2$ or $A \times V_1 = 2A\sqrt{\frac{g\ell}{2}}$ or $V_1 = \sqrt{2g\ell}$

$\frac{1}{2}\rho\left[V_1^2 - V_2^2\right] = \rho g\ell \sin\theta$

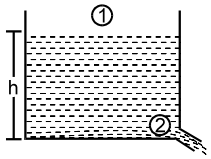
$\Rightarrow \qquad \frac{1}{2}\left[2g\ell - \frac{g\ell}{2}\right] = g\ell \sin\theta$

on solving $\sin\theta \frac{3}{4}$.

15. The velocity of fluid at the hole is $V_2 = \sqrt{\dfrac{2gh}{1+(a^2/A^2)}}$

Using continuity equation at the two cross-sections (1) and (2) :

$V_1 A = V_2 a \quad \Rightarrow \quad V_1 = \dfrac{a}{A} V_2$



$\Rightarrow$ Acceleration (of top surface) $= -V_1 \dfrac{dV_1}{dh}$

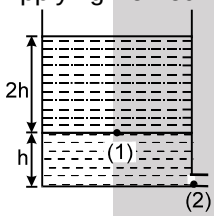
$= -\dfrac{a}{A} V_2 \dfrac{d}{dh}\left(\dfrac{a}{A} V_2\right)$

$a_1 = -\dfrac{a^2}{A^2} V_2 \dfrac{dV_2}{dh} = -\dfrac{a^2}{A^2}\sqrt{2gh}\sqrt{2g}\cdot\dfrac{1}{2\sqrt{h}} \Rightarrow a_1 = \dfrac{-ga^2}{A^2}$

16. Pressure at (1) :

$P_1 = P_{atm} + \rho g (2h)$

Applying Bernoulli's theorum between points (1) and (2)



$[P_{atm} + 2\rho g h] + \rho g(2h) + 1/2\,(2\rho)\,(0)^2$

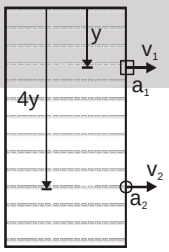
$= P_{atm} + (2\rho)\, g\,(0) + 1/2\,(2\rho)\, v^2 \Rightarrow \quad v = 2\sqrt{gh}$ **Ans.**

17. by $A_1 V_1 = A_2 V_2$

$\left(\dfrac{\pi D_1^2}{4}\right) V_1 = \left(\dfrac{\pi D_2^2}{4}\right) V_2; \quad V_1 = 4V_2$

18. Velocity of efflux at a depth h is given by $V = \sqrt{2gh}$

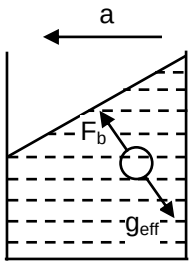
Volume of water following out per second from both the holes are equal



$\therefore \quad a_1V_1 = a_2V_2$

or $(L^2)\sqrt{2g(y)} = \pi R^2 \sqrt{2g(4y)}$ or $R = \dfrac{L}{\sqrt{2\pi}}$

19.



w.r.t. container

20. $\frac{\rho A}{\rho \ell} = \frac{1}{2} \quad \Rightarrow \quad \frac{\rho A}{\rho B} = \frac{1}{3}$

$\rho_A Vg + \rho_B Vg = \rho \ell_2 Vg$

$\rho_A + \rho_B = 2\rho_e$

$\rho_B = 3\rho_A$

$\rho_\ell = 2\rho_A$

$\rho_\ell : \rho_A : \rho_B = 2 : 1 : 3$

21. $\frac{4}{3}\pi\left(1000r^3 - 729r^3\right)\rho = \frac{2}{3}\pi 1000r^3 0.8\rho\omega$

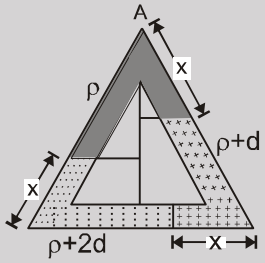
$271\rho = 4\omega\rho_\omega$

$\rho = 1.48\rho_\omega = 1.48$ gm/cm^3

PART - II

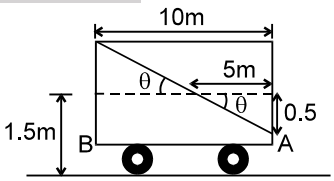
1. $(\rho + 2d)\ x\sin 60 + \rho(\ell - x)\ \sin 60^o = (\rho + d)\ (\ell - x) \sin 60^o + \rho x \sin 60^o$

On solving



$x = \frac{\ell}{3} = \frac{3}{3} = 1$

2. $\tan\theta = \frac{0.5}{5} = \frac{1}{10}$



$\frac{a}{g} = \frac{1}{10}$

$a = 1$ m/s^2

$v = u + at$

$10 = 0 + 1 \times t$

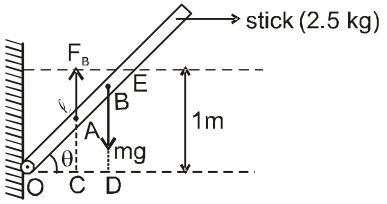
$t = 10$ second.

3. OE = ℓ

$\ell = \dfrac{1}{\sin\theta}$ ...(i)

OD = 2 cos θ ...(ii)

$OC = \dfrac{\ell}{2}\cos\theta = \dfrac{\cos\theta}{2\sin\theta}$ ...(iii)



$F_B = \rho_w A\ell g$

torque about O

mg (OD) = $\rho_w A\ell g$(OC)

on solving

$\sin^2\theta = \dfrac{1}{4}$

$\theta = 30^o$ **Ans.**

For height $\sin^2\theta < \dfrac{h^2}{4}$ (h is depth of water)

For $\theta = 90^o$, h > 2

4. Breaking force = Breaking stress × area of wire

$= 7 \times 10^6 \times 10^{-6} = 7N$

weight of block = vol × density × g

$= 10^{-3} \times (1.5 \times 10^3) \times 10 = 15$ N.

The upthrust when the liquid level has descended by x cm.

$= 100\,(10 - x) \cdot 10^{-6} \cdot 10^3 \cdot 10$

= 10 – x Newton.

$\therefore$ Net downward force on the block = 15 – (10 – x) = 5 + x = Tension T

$\therefore$ The wire will break when 5 + x = 7.

i.e. when x = 2cm

Let the level descends by 2 cm in t time

then $2t = (200 - 100).2$

t = 100 sec. .....Ans.

$20x = 100 \quad \therefore x = 5$

5. $\rho aV^2 = \mu N$

$1000 \times a\,(2g \times 2) = 0.4 \times 100 \times 10$

$1000 \times \pi \dfrac{d^2}{4} \times 4g = 400$ or $d = \dfrac{0.2}{\sqrt{\pi}}$

6. vt = 80 m ...(i)

$t = \sqrt{\dfrac{2h}{g}} = \sqrt{\dfrac{2\times 20}{10}} = 2$ second ....(ii)

from (i) and (ii) equ.

v = 40 m/s. ....(iii)

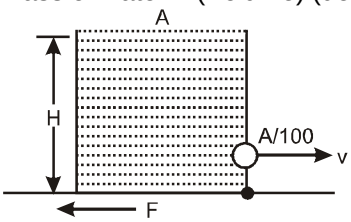
$P_0 + \rho gh = P_{atm} + \dfrac{1}{2}\rho v^2$ ...(iv)

on solving (iv) equation

$P_0 = 2.78 \times 10^5\ Nm^{-2} = (1.39)\alpha \times 10^5$

$\alpha = 2$ **Ans.**

7. Mass of water = (Volume) (density)



$\therefore$ $m_0 = (AH)\,r$

$\therefore$ $H = \dfrac{m_0}{A\rho}$

Velocity of efflux, $V = \sqrt{2gH} = \sqrt{2g\dfrac{m_0}{A\rho}} = \sqrt{\dfrac{2m_0g}{A\rho}}$

Thrust force on the container due to draining out of liquid from the bottom is given by,
F = (density of liquid) (area of hole) (velocity of effux)^2
($F = \rho aV^2$)

$F = \rho(A/100)V^2 = \rho(A/100)\left(\dfrac{2m_0g}{A\rho}\right)$

$F = \dfrac{m_0g}{50}$

$\therefore$ Acceleration of the container, $a = F/m_0 = g/50 = \dfrac{1}{5} = 0.2\ m/s^2$

So $x = 2$ **Ans.**

8. Given : $A_1 = 4 \times 10^{-3}\ m^2$, $A_2 = 8 \times 10^{-3}\ m^2$
$h_1 = 2m$, $h_2 = 5m$
$v_1 = 1\ m/s$ and $\rho = 10^3\ kg/m^3$
From continuity equation, we have

$A_1 v_1 = A_2 v_2$ or $v_2 = \left(\dfrac{A_1}{A_2}\right) v_1$

or $v_2 = \left(\dfrac{4\times10^{-3}}{8\times10^{-3}}\right)(1\ m/s)$

$v_2 = \dfrac{1}{2}\ m/s$

Applying Beronulli`s equation at section 1 and 2

$P_1 + \dfrac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \dfrac{1}{2}\rho v_2^2 + \rho g h_2$

or $P_1 - P_2 = \rho g\,(h_2 - h_1) + \dfrac{1}{2}\rho\left(v_2^2 - v_1^2\right)$ ...(2)

(i) Work done per unit volume by the pressure as the fluid flows from P to Q.

$W_1 = P_1 - P_2$

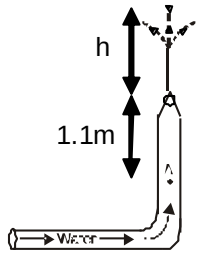
$= \rho g\,(h_2 - h_1) + \dfrac{1}{2}\rho\left(v_2^2 - v_1^2\right)$ [From Eq. (1)]

$= \left\{(10^3)(9.8)(5-2) + \dfrac{1}{2}(10^3)\left(\dfrac{1}{4} - 1\right)\right\} J/m^3 = [29400 - 375]\ J/m^3 = 29025\ J/m^3$

$1161\ \alpha = 29025\ J/m^3$

$\alpha = 25$

9.



$A_1v_1 = A_2v_2$

$Av_A = \frac{A}{4}v_n$

$v_n = 4v_A$ ....(i)

$P_A + \frac{1}{2}\rho v_A^2 = \frac{1}{2}\rho v_n^2 + \rho g\hbar$ ...(ii)

$v_n^2 / 2g = \hbar$ ....(iii)

On solving (i), (ii) & (iii) equation we get $\hbar = 3.2$ m **Ans.**

PART - III

1. PV = constant
(Assumed isothermal process)

2. In a static fluid, pressure remains same at the same level, ie, pressure do not vary with x-coordinate.
Hence (C).

3. $P = \rho(2h)g$

$\frac{F}{A_2} = \rho\ (2h)\ g$

$F_{base} = 2h\ \rho\ g\ A_2$

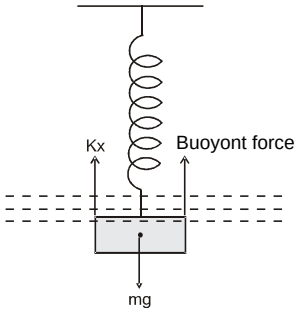
$F_{wall} = h\ \rho\ g\ [A_2 - A_1]$, at the level x

4. Assuming that the block is completely submerged in water, then
$F_b = 1000 > mg(920)$ So, not possible
Let complete in oil
$F_b = (0.6)\ (4)\ (1000 + (1)\ (6)\ (100) = 840$
$F_b < mg$ So, not possible
So, let 'x' part in oil and remaining in water
$920 = [(1)\ (10 - x) + (0.6)\ (x)]\ 100$
$9.2 = 10 - x + 0.6x$
$0.4x = 0.8$
$x = 2$ cm.

5. $F_b = v\rho_{liq}g$
'g' is different on moon and on the earth.
Hence only (iii) is a correct statement.
Hence (D).

6. ℓ will decrease because the block moves up. h will decrease because the coin will displace the volume of water (V_1) equal to its own volume when it is in the water whereas when it is on the block it will displace the volume of water (V_2) whose weight is equal to weight of coin and since density of coin is greater than the density of water $V_1 < V_2$.

7. $Kx = 10 \times 10 - \frac{1000}{2000} \times 10 \times 10$



$Kx = 50$ N

$U_{stored} = \frac{1}{2} \times (100) \left(\frac{50}{100}\right)^2 = \frac{1}{2} \times \frac{2500}{100} = 12.5$ J

8. $x = 2\sqrt{H(H-h)}$

$x_1 = 2\sqrt{70 \times 20}$

$x_2 = 2\sqrt{60 \times 30}$

$x_3 = 2\sqrt{40 \times 50}$

$x_4 = 2\sqrt{50 \times 40}$ or $x_3 = x_4$ = maximum

PART - IV

1. As, $dm = A\rho_W v\, dt$

$\Rightarrow \quad \frac{dm}{dt} = A\rho_W v$

$\Rightarrow \quad \frac{dm}{dt} = V\rho_W \pi \frac{D^2}{4}$

where 'D' is the diameter of stream.

2. $V_1A_1 = V_2A_2$

$\frac{\pi v_0 D_0^{\,2}}{4} = \frac{\pi v D^2}{4} \quad \Rightarrow \quad D = D_0\sqrt{\frac{v_0}{v}}$.

3. $v = \sqrt{u^2 + 2gh} = \sqrt{2g(b+x)}$.

4. Applying continuity equation at points with diameter D_0 & D :

$= \sqrt{2gb}.\left[\frac{\pi.D_0^{\,2}}{4}\right] = \sqrt{2g(b+x)}\left[\frac{\pi D^2}{4}\right]$

$\Rightarrow \quad D = D_0\left[\frac{b}{b+x}\right]^{1/4}$

5. Solving the preceding formula for the tank height h gives :
$h = x(D/D_0)^4/(1 - (D/D_0)^4) = x\,D^4 / (D_0^4 - D^4)$
Substituting the given parameter values gives
$h = (0.3)\,(0.009^4) / (0.01^4 - 0.009^4) = 0.57$ m
So the height of the water above the tap is 0.57 m or 57 cm.

6. Apparent weight ($W_{app.}$) = $W - V\rho_\ell g$
Since, $W_{app.\,(Ram)} > W_{app.\,(Shyam)}$
$\Rightarrow \quad W_{(Ram)} > W_{(Shyam)}$
Therefore, from given passage shyam has more fat than Ram.

7. $V_1 > V_2 \Rightarrow W_{app.\,(1)} < W_{app.\,(2)}$
(Since $W_{app.} = W - V\rho_\ell g$)
Hence (B)

8. $\rho_{Salt\ water} > \rho_{Fresh\ water} \Rightarrow W_{app.\,(s)} < W_{app.\,(F)}$
Hence (A)

9. Let 'V' be the total volume of the person

Then ; $\left(\frac{V}{4}\right)(0.4 \times 10^3) + \left(\frac{3}{4}V\right)\left(\frac{4}{3}\times 10^3\right) = 165$

$\Rightarrow \quad V = \frac{165}{1100} = 0.15\ m^3$

Reading on spring balance under water is :
$W_{app} = [165 \times 10] - 0.15\,[10^3]\,[10] = 150\ N$

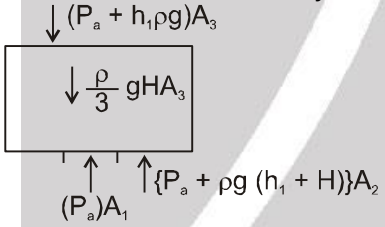
10. Just after the string is cut :

$a = \frac{150}{165} = 0.91\ m/s^2$ **Ans.**

EXERCISE-3

PART - I

1. $A_1 = \pi r^2$ = area of base of cylinder in air
$A_2 = 3\pi r^2$ = area of base of cylinder in water
$A_3 = 4\pi r^2$ = cross-section area of cylinder



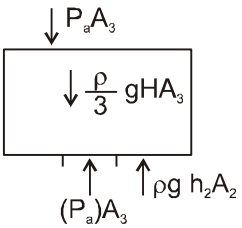
from the equilibrium of block (see diagram)
Equating the forces, we get

$(P_a + \rho g h_1)A_3 + \frac{\rho}{3} gH\ A_3 = (P_a)A_1 + [P_a + \rho g(h_1 + H)]A_2$

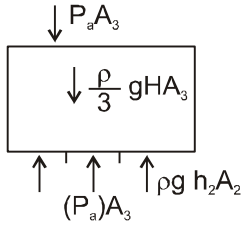
On solving

$h_1 = \frac{5}{3}H$

2. (A) $P_aA_3 + \frac{\rho}{3}A_3Hg = P_aA_3 + \rho g h_2 A_2$



from the equilibrium of block (see diagram)

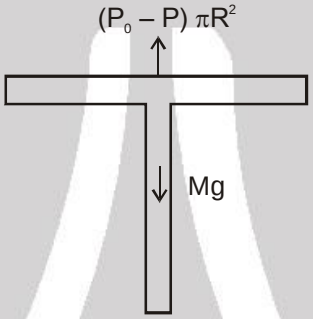


$h_2 = 4H/9.$

3. For $h_2 < 4h/9$ cylinder does not move up because further bouyant force decreases while the weight of block remains same.

4. Since it is open from the top, the pressure will be P_0

5. Resultant force on the piston is zero (Let pressure in air be P)
From the equilibrium of the piston



$(P_0 - P)\,\pi R^2 = Mg$

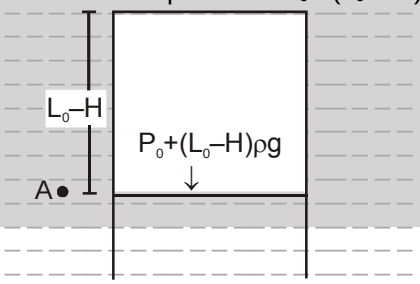
$$P = P_0 - \frac{Mg}{\pi R^2}$$

From the conservation of moles of air : $P_1V_1 = P_2V_2$, it follows that

$P_0 \cdot 2L = Px$

$$\therefore x = \frac{P_0.2L}{P} = \frac{P_0 2L}{P_0 - \frac{Mg}{\pi R^2}}$$

6. Pressure in air inside cylinder = Pressure at point A = $P_0 + (L_0 - H)\,\rho g$



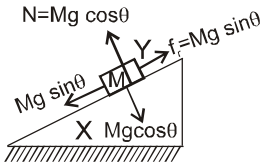
PV = constant in the air inside the cylinder

$\therefore P_0L_0 = [P_0 + (L_0 - H)\rho g]\,(L_0 - H)$

$P_0\,(L_0 - H) + \rho g\,(L_0 - H)^2 - P_0L_0 = 0$

7. As the stream falls down, its speed will increase and cross-section area will decrease.
Thus it will become narrow.
Similarly as the stream will go up, speed will decrease and cross-section area will increase.
Thus it will become broader.
Hence statement-1 is correct and statement-2 is correct explanation also.
Ans. (A)

8. (p)



Net force on Y due to X = $\sqrt{(mg\cos\theta)^2+(mg\sin\theta)^2}$ = mg

(B) As the inclined is fixed. So, gravitational P.E. of X is constant

(C) As K.E. is constant and P.E. of Y is decreasing. So mechanical energy of (X + Y) is decreasing.

(q)

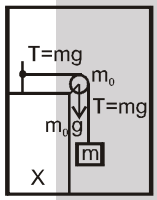
(A) force on Y due to X will be greater than mg which is equal to (Mg + repulsion force)

(B) As the system is moving up, P.E. of X is increasing.

(C) Mechanical energy of (X + Y) is increasing

(D) Torque of the weight of Y about point P = 0

(r)



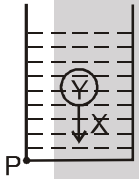
(A) force on Y due to X = $\sqrt{[(m+m_0)g]^2+(mg)^2}$

(B) As the system moves down, gravitational P.E. of X decreases

(C) As the system moves down, total mechanical energy of (X + Y) also decreases

(D) $\tau_P \neq 0$

(s)



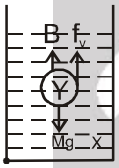
(A) force on Y due to X = Buoyancy force which is less than mg

(B) As the sphere moves down, that volume of water comes up, so gravitational P.E. of X increases.

(C) As there is no non–conservative force, so total mechanical energy of X + Y remains conserved.

(D) $\tau_p \neq 0$

(t)



(A) As the sphere is moving with constant velocity

$B + f_v = Mg$

so force on Y due to X is $B + f_v = mg$

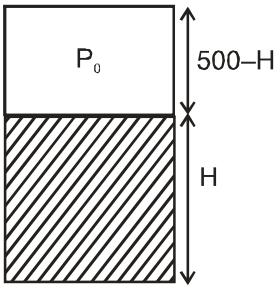
(B) As the sphere moves down, that volume of water comes up, so gravitational P.E. of X will increase

(C) Increase in mechanical energy

w_{fr} = –ve

(D) $\tau_p \neq 0$

9.



P_0 = atmospheric pressure

$P + 200 \times 10^{-3} \times 1000 \times 10 = P_0$ ....(1)

$P_0 (500 - H) = P \,.\, (300 \text{ mm})$

$$\Rightarrow \quad P = \frac{P_0(500-H)\ \text{mm}}{300\ \text{mm}} \qquad \text{....(2)}$$

from (1) and (2)

$$\frac{P_0(500-H)}{300} + 2000 = P_0$$

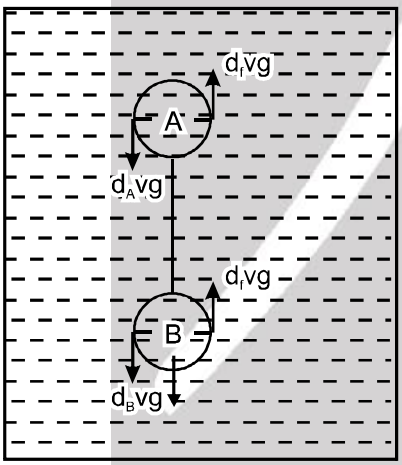
$$\frac{10^5(500-H)}{300} + 2000 = 10^5$$

$\Rightarrow \quad 5 \times 10^7 - H \times 10^5 + 6 = 300$

$\Rightarrow \quad H = 206$ mm

fall in height = 6 mm **Ans. 6**

10.



For equilibrium

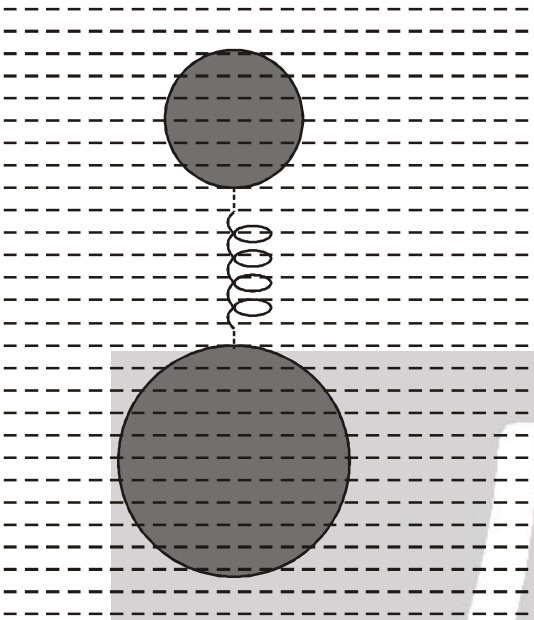
$d_A Vg + d_B Vg = d_F Vg + d_F Vg$

$\Rightarrow \ d_F = \frac{d_A + d_B}{2}$ $\Rightarrow$ Option (D) is correct

to keep the string tight

$d_B > d_F$ and $d_A < d_F$

11.



On small sphere

$$\frac{4}{3}\pi R^3(\rho)g + kx = \frac{4}{3}\pi R^3(2\rho)g \quad \text{....(i)}$$

on second sphere (large)

$$\frac{4}{3}\pi R^3(3\rho)g = \frac{4}{3}\pi R^3(2\rho)g + kx \quad \text{....(ii)}$$

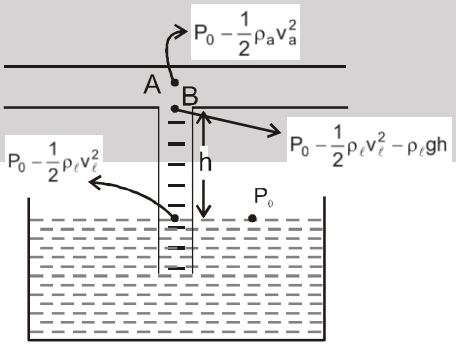
by equation (i) and (ii)

$$x = \frac{4\pi R^3\rho g}{3k}$$

12. $A_1 V_1 = A_2V_2$ $\quad A_1 = 400\ A_2$

$400\ (5 \times 10^{-3}) = V_2 \quad \Rightarrow \quad \boxed{V_2 = 2 \quad m/s} \quad$ (C)

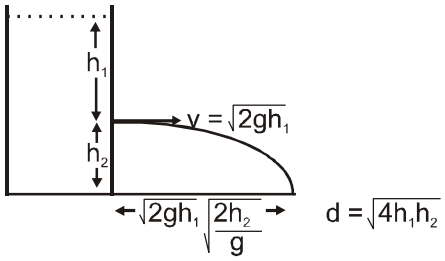
13. Pressure at A and B will be same



$$P_0 - \frac{1}{2}\rho_a v_a^2 = P_0 - \frac{1}{2}\rho_\ell v_\ell^2 - \rho_\ell gh$$

$$v_\ell = \sqrt{\frac{\rho_a}{\rho_\ell}}v_a - 2gh$$

14. Match the column
When lift is at rest :



(P) $g_{eff} > g$ $d = \sqrt{4h_1h_2} = 1.2$ m

(Q) $g_{eff} < g$ $d = \sqrt{4h_1h_2} = 1.2$ m

(R) $g_{eff} = g$ $d = \sqrt{4h_1h_2} = 1.2$ m

(S) $g_{eff} = o$ No water leaks out of the jar.

PART - II

1. Since solid ball floats in between the two liquids hence $\rho_1 < \rho_3 < \rho_2$

2. For equilibrium, weight should be balanced by buoyant force.
density of oil < density of water
and ball should be in between oil and water.

3. Diameter = 8×10^{-3} m
v = 0.4 m/s

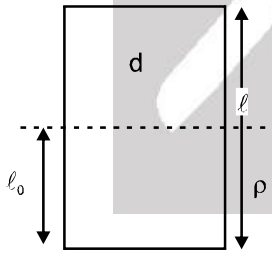
$v = \sqrt{u^2 + 2gh} = \sqrt{(0.4)^2 + 2 \times 10 \times 0.2} = 2$ m/s

$A_1v_1 = A_2v_2$

$$\pi\left(\frac{8\times10^{-3}}{2}\right)^2 \times 0.4 = \pi \times \frac{d^2}{4} \times 2$$

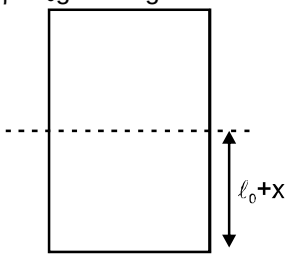
$d \approx 3.6 \times 10^{-3}$ m.

4.



At equilibrium
$F_b = mg$

$\rho A\ell_0 g = dA\ell g$ .....(i)



Restoring force,
$F = mg - F_b'$

$F = mg - \rho A(\ell_0 + x)g$

$dA\ell a = dA\ell g - \rho A\ell_0 g - \rho g A x$

$a = -\frac{\rho g}{d\ell} x$

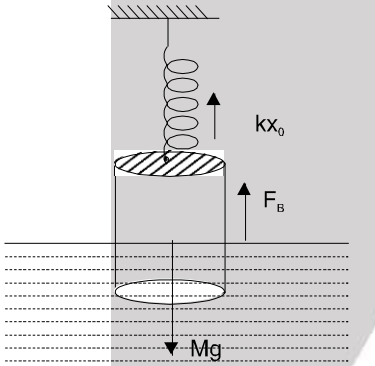
$\omega = \sqrt{\frac{\rho g}{d\ell}}$

$T = 2\pi\sqrt{\frac{\ell d}{\rho g}}$ .....(i)

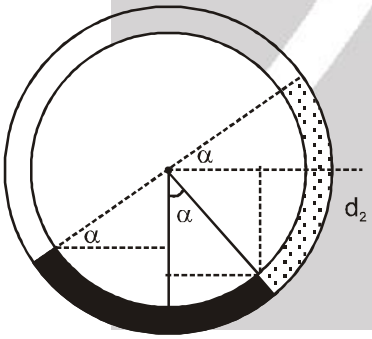
5. $kx_0 + F_B = mg$

$kx_0 + \frac{L}{2}\sigma Ag = Mg$

$x_0 = \frac{Mg - \frac{\sigma LAg}{2}}{k} = \frac{Mg}{k}\left(1 - \frac{\sigma LA}{2M}\right)$



6.



$R\sin\alpha\, d_2 + R\cos\alpha\, d_2 + R(1 - \cos\alpha)d_1$

$= R(1 - \sin\alpha)\, d_1$

$(\sin\alpha + \cos\alpha)\, d_2 = d_1(\cos\alpha - \sin\alpha)$

$\Rightarrow \quad \frac{d_1}{d_2} = \frac{1 + \tan\alpha}{1 - \tan\alpha}$

HIGH LEVEL PROBLEMS (HLP)

1. In elastic collision with the surface, direction of velocity is reversed but its magnitude remains the same. Therefore, time of fall = time of rise.
or time of fall = $t_{1/2}$
Hence, velocity of the ball just before it collides with liquid is

$$v = g\frac{t_1}{2} \qquad ....(1)$$

Retardation inside the liquid

$$a = \frac{\text{upthrust} - \text{weight}}{\text{mass}}$$

$$= \frac{Vd_Lg - Vdg}{Vd} = \left(\frac{d_Lg - d}{d}\right) g \text{(V = volume of ball)} \qquad ....(2)$$

Time taken to come to rest under this retardation will be

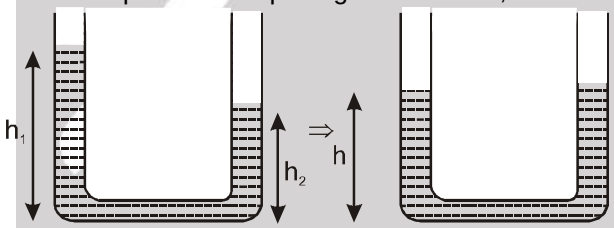
$$t = \frac{v}{a} = \frac{gt_1}{2a} = \frac{gt_1}{2\left(\frac{d_L - d}{d}\right) g} = \frac{dt_1}{2(d_L - d)}$$

Same will be the time to come back on the liquid surface.
Therefore,
(a) t_2 = time the ball takes to came back to the position from where it was released

$$= t_1 + 2t = t_1 + \frac{dt_1}{d_L - d} = t_1 \left[1 + \frac{d}{d_L - d}\right] \quad \text{or} \quad t_2 = \frac{t_1 d_L}{d_L - d}$$

(b) The motion of the ball is periodic but not simple harmonic because the acceleration of the ball is g in air and $\left(\frac{d_L - d}{d}\right) g$ inside the liquid which is not proportional to the displacement, which is necessary and sufficient condition for SHM.
(c) When $d_L = d$, retardation or acceleration inside the liquid becomes zero (upthrust = weight). Therefore, the ball will continue to move with constant velocity $v = gt_1 / 2$ inside the liquid.

2. Let h be level in equilibrium. Equating the volumes, we have



$$Ah_1 + Ah_2 = 2Ah$$

$$\therefore \quad h = \left(\frac{h_1 + h_2}{2}\right)$$

Work done by gravity = $U_i - U_f$

$$W = \left(m_1 g\frac{h_1}{2} + m_2 g\frac{h_2}{2}\right) - (m_1 + m_2)g\frac{h}{2}$$

$$= \frac{Ah_1\rho g h_1}{2} + \frac{Ah_2\rho g h_2}{2} - [Ah_1\rho + Ah_2\rho]g\left(\frac{h_1 + h_2}{4}\right)$$

Simplifying this, we get

$$W = \frac{\rho A g}{4}(h_1 - h_2)^2$$

3.

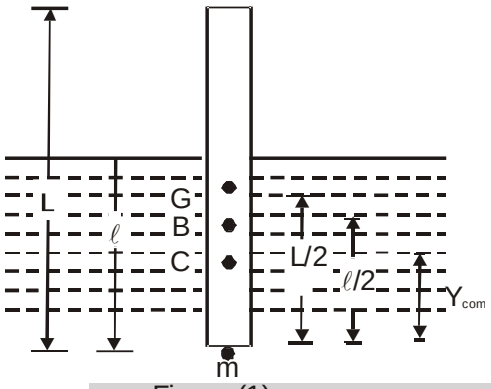


Figure (1)

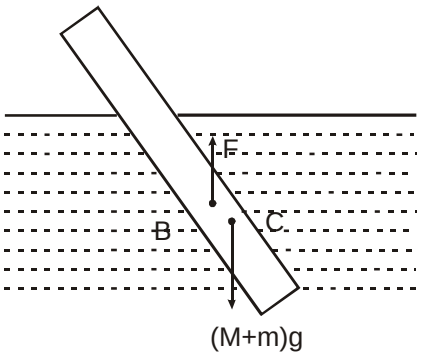


Figure (2)

Let M = Mass of stick = $\pi R^2 \rho L$

ℓ = Immersed length of the rod

G = COM of rod

B = Centre of buoyant force (F)

C = COM of rod + mass (m)

Y_{com} = Distance of C from bottom of the rod

Mass m should be attached to the lower end because otherwise B will be below G and C will be above G and the torque of the couple of two equal and opposite forces F and (M + m) g will be counter clockwise on displacing the rotational equilibrium. See the figure 3 given alongside.

For vertical equilibrium

Mg + mg = F (upthrust)

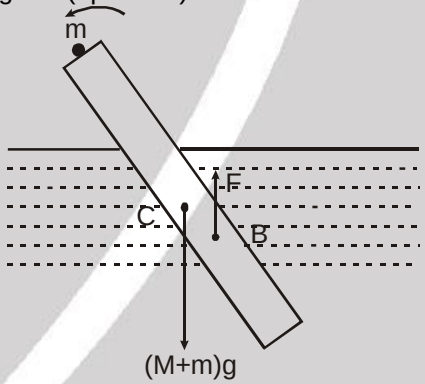


Figure (3)

or $(\pi R^2 Lg\rho) + mg = (\pi R^2 \ell)\sigma g$

$\therefore \quad \ell = \left(\dfrac{\pi R^2 L\rho + m}{\pi R^2 \sigma}\right)$ .....(1)

Position of COM (of rod + m) from bottom

$$Y_{com} = \frac{M\frac{L}{2}}{M+m} = \frac{(\pi R^2 L\rho)\frac{L}{2}}{(\pi R^2 L\rho)+m} \qquad .....(2)$$

Centre of buoyancy (B) is at a height of $\dfrac{\ell}{2}$ from the bottom.

We can see from figure (2) that for rotational equilibrium of the rod, B should either lie above C or at the same level of B.

Therefore $\dfrac{\ell}{2} \geq Y_{COM}$ or $\dfrac{\pi R^2 L\rho + m}{2\pi R^2 \sigma} \geq \dfrac{\left(\pi R^2 L\rho\right)\frac{L}{2}}{\left(\pi R^2 L\rho\right)+m}$

or $m + \pi R^2 L\rho \geq \pi R^2 L \sqrt{\rho\sigma}$ or $m \geq \pi R^2 L\ (\sqrt{\rho\sigma} - \rho)$

$\therefore$ Minimum value of m is $\pi R^2 L\ (\sqrt{\rho\sigma} - \rho)$ **Ans.**

4. (a) (i) Considering vertical equilibrium of cylinder :
Weight of cylinder = upthrust due to upper liquid
+ upthrust due to lower liquid

Note that h_1 and $h_2 \neq \frac{H}{2}$

$\therefore \left(\frac{A}{5}\right)$ (L) D.g = $\left(\frac{A}{5}\right)\left(\frac{3L}{4}\right)$ (d) g + $\left(\frac{A}{5}\right)\left(\frac{L}{4}\right)$ (2d) (g)

$\therefore$ $D = \left(\frac{3}{4}\right)d + \left(\frac{1}{4}\right)(2d)$

$D = \frac{5}{4}d$

(ii) Considering vertical equilibrium of two liquids and the cylinder.
$(P - P_0)$ A = weight of two liquids + weight of cylinder

$\therefore$ $P = P_0 + \frac{\text{weight of two liquids} + \text{weight of cylinder}}{A}$ ...(1)

Now , weight of cylinder

$= \left(\frac{A}{5}\right)$ (L) (D) (g) $= \left(\frac{A}{5}Lg\right)\left(\frac{A}{5}d\right)$

$= \frac{ALdg}{4}$

Weight of upper liquid $= \left(\frac{H}{2}Adg\right)$ and

Weight of lower liquid $= \frac{H}{2}$ A (2d)g = HAgd

$\therefore$ Total weight of two liquids = $\frac{3}{2}$HAdg

$\therefore$ From Eq. (1) pressure at the bottom of the container will be

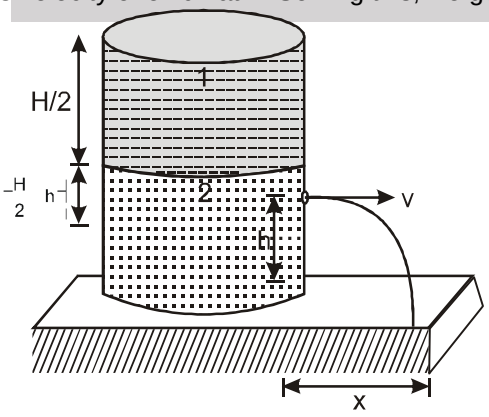
$$P = P_0 + \frac{\left(\frac{3}{2}\right)HAdg + \frac{ALdg}{4}}{A}$$

or $P = P_0 + \frac{dg(6H + L)}{4}$

(b) (i) Applying Bernoulli`s theorem at 1 and 2

$P_0 + dg\left(\frac{H}{2}\right) + 2dg\left(\frac{H}{2} - h\right) = P_0 + \frac{1}{2}(2d)\,v^2$

Here, v is velocity of efflux at 2. Solving this, we get



$v = \sqrt{(3H - 4h)\frac{g}{2}}$

(ii) Time taken to reach the liquid to the bottom will be

$$t = \sqrt{2h/g}$$

\ Horizontal distance x travelled by the liquid is

$$x = vt = \sqrt{\left(3H - 4h\right)\frac{g}{2}}\left(\sqrt{\frac{2h}{g}}\right)$$

$$x = \sqrt{h(3H - 4h)}$$

(iii) For x to be maximum

$$\frac{dx}{dh} = 0$$

or $$\frac{1}{2\sqrt{h(3H-4h)}}\ (3H - 8h) = 0$$

or $$h = \frac{3H}{8}$$

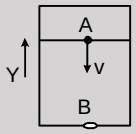
Therefore, x will be maximum at $h = \frac{3H}{8}$.

The maximum value of x will be

$$x_m = \sqrt{\left(\frac{3H}{8}\right)3H - 4\left(\frac{3H}{8}\right)}$$

$$x_m = \frac{3}{4}H$$

5. Let the velocity of efflux of mercury coming out of hole be ν at an instant mercury level in container is y. At same instant the speed of top surface of fluids v. from equation of continuity



$$\frac{S}{n}\ v = Sv \qquad ....(1)$$

$\because$ $\frac{S}{n} << S$ $\quad\therefore\quad \nu >> v$

applying Bernoulli's theorem between A and B

$$Yeg + \frac{1}{2}\ ev^2 = P_{atm} + \frac{1}{2}ev^2$$

$\because$ $v << \nu$ higher powers of v can be neglected and $P_{atm} = h_0 eg$

$\therefore$ $\nu = \sqrt{2g(Y - h_0)}$ .....(2)

Hence mercury flows out of both till $y = h_0$.
from equation (1)

$$v = -\frac{dy}{dt} = \frac{1}{h}\nu = \frac{1}{h}\sqrt{2g(Y - h_0)}$$

or $$= \frac{-dy}{\sqrt{2gY - h_0}}\ \frac{1}{h}\ dt$$

integrating between limits

at $l = 0$ $y = n$ and $t = T$, $Y = h_0$; $-\int_n^{h_0} \frac{dy}{\sqrt{2g\ (Y - h_0)}} = \frac{1}{n}\int_0^T dt$; $t = n\sqrt{\frac{2}{g}(h - h_0)}$

6. Bernoulli's equation between A and B gives

$$\frac{p_A}{\rho_a} = \frac{p_B}{\rho_a} + \frac{v^2}{2} \Rightarrow \qquad v^2 = 2\left[\frac{p_A - p_B}{\rho_a}\right]$$

Also equating pressures at horizontal level of E

$$p_A + \rho_a gy + \rho_a gh = p_B + \rho_a gy' + \rho_a gy + \rho_m gh.$$

$$\Rightarrow \quad p_A + \rho_a gh = p_B + \rho_m gh \qquad [\because y' 0]$$

$$p_A - p_B = (\rho_m - \rho_a)\, gh.$$

$$v^2 = \frac{2(\rho_m - \rho_a)gh}{\rho_a}$$

$$\mathbf{v = \sqrt{\frac{2(\rho_m - \rho_a)gh}{\rho_a}}} \qquad \textbf{Ans.}$$

7. Weight of sphere + chain = $(m + \lambda h)g$

Buoyant force = $(3m + \frac{\lambda h}{7})\, g$

for equilibrium, weight = Buoyant force or , $m + \lambda h = 3m + \frac{\lambda h}{7}$ or $h = \frac{7m}{3\lambda}$

8. Apply Bernolli's equation b/w point P & D

$$P_{atm} + \rho g h_1 + \frac{1}{2}\rho V_p^2 = P_{atm} + \rho g(o) + \frac{1}{2}\rho v_D^2$$

[Assume zero level D]

$V_p \approx 0$ Area of cross section is very large

$V_D = \sqrt{2gh_1}$

Since area at C is half than area at D
so according to continuity equation

$$V_C A_C = V_D A_D \quad \Rightarrow \quad V_C = 2V_D = 2\sqrt{2gh_1}$$

Now for point P & C according to Bernolli's equation.

$$P_p + \rho g h_1 + \frac{1}{2}\rho V_p^2 = P_C + \rho g(0) + \frac{1}{2}\rho . V_C^2$$

$V_p \approx 0$

$$\Rightarrow \quad P_{atm} + \rho g h_1 = P_C + \frac{1}{2}\rho . (2\sqrt{2gh_1})^2$$

$$\Rightarrow \quad P_{atm} = P_C + 3\rho g h_1 \qquad .....(i)$$

for point E

$$P_E = P_C + \rho g h_2 = P_{atm} \qquad .....(ii)$$

from (i) & (ii)

$$3\rho g h_1 = \rho g h_2$$

$$\Rightarrow \quad \mathbf{h_2 = 3h_1} \qquad \textbf{Ans.}$$